

1. Graph the function $r = 3(1 - \sin(\theta))$.
2. Set up an integral to find the area enclosed by $r = 3(1 - \sin(\theta))$.
3. Find the area enclosed by three leaves of the four-leafed rose $r = 5 \cos(2\theta)$.
4. Evaluate $\int \cos^2(3x) \, dx$.

5. Set up an integral to find the length of the curve $y = 4 \ln x$, from $x = 1$ to $x = 3$
- (a) By integrating with respect to x .

 - (b) By integrating with respect to y .
6. Set up an integral to find the area of the surface generated by rotating the curve $y = 4 \ln x$, from $x = 1$ to $x = 3$
- (a) About the x -axis.

 - (b) About the line $y = -3$.

 - (c) About the line $x = -3$.